1. Which term refers to the farming of aquatic organisms under controlled conditions for commercial
purposes?
A. Aquaculture
B. Pisciculture
C. Hydrology
D. Holticulture
Answer: A
2. Which discipline of biology studies the physical, chemical, and biological properties of drugs from
natural sources?
A. Pharmacology
B. Pharmacognosy
C. Biochemistry
D. Microbiology
Answer: B
3. Biology being the study of living things helps us to understand every _____ .
A. living plants
B. living animals
C. living organisms
D. environment
Answer: C
4. Jute fiber, used in making textiles and gunny bags, is derived from which plant species?
A. Gossypium
B. Cinchona
C. Catharanthus
D. Corchorus
Answer: D
5. Gene expression can produce some _____, that are not translated into proteins.
A. tRNA
B. rRNA
C. RNAs
D. mRNA
Answer: C
6. What is the main purpose of aquaculture?
A. Wild stock population enhancement
B. Food production
C. Habitat restoration
D. Recreation
Answer: B
7. Which of the following advances have brought improvements in quality of life?
A. all of these
B. medicine
C. agriculture

D. biotechnology
Answer: A

8. Feedback is a type of _____ common to life at all levels.
A. changes
B. alternation
C. transformation
D. regulation
Answer: D
9. Mark the incorrect match.
A. Hirudomedicinalis – Leech
B. Luciliasericata – Green bottle fly
C. Bombyx mori – Silkworm
D. Xyliadolabriformis– Spider
Answer: D
10. What is a defining characteristic of eukaryotic cells?
A. Lack of a nucleus
B. Presence of membrane-enclosed organelles
C. Small size and simple structure
D. Absence of genetic material
Answer: B
11. Producers convert energy from _____ to chemical energy.
A. plants
B. environment
C. sunlight
D. food
Answer: C
12. Which of the following paring is correct?
A. Pisciculture - fish
B. Sericulture – Seaweed
C. Mariculture – Medicinal plants
D. Apiculture – Fruits
Answer: A
13. Maize seeds should not be sown more than _____ depth of soil.
A. 5-7 cm
B. 4-6 cm
C. 5-4 cm
D. 5-6 cm
Answer: D
14. What is the scientific name pigeonpea ?
A. Vigna mungo
B. Sesamum indicum
C. Cajanus cajan
D. Arachis hypogaea

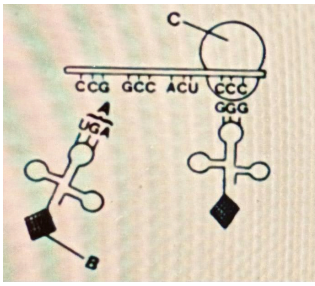
Answer: C
15. Which nitrogenous bases have a single ring in their chemical structure:
A. adenine and guanine
B. cytosine and thymine
C. adenine and thymine
D. cytosine and thymine
Answer: D

16. One of the DNA strands which occur Okazaki fragments is called:
A. Lagging strand
B. Leading strand
C. Continuous strand
D. Parent strand
Answer: A
17. Genes are isolated from human DNA using ……1…… enzymes. A bacterial plasmid is cut with the
same enzyme forming ……2…… . The human DNA is inserted into the bacterial plasmid using the
enzyme ……3…… forming a ……4…… plasmid. Which row correctly completes gaps 1, 2, 3 and 4?

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| A | ligase | Sticky ends | protease | restriction |
| B | recombinant | New DNA | ligase | daughter |
| C | restriction | Daughter plasmids | ligase | diploid |
| D | restriction | Sticky ends | ligase | recombinant |

Answer: D

18. The main enzyme in the DNA replication is:
A. DNA polymerase
B. DNA ligase
C. DNA primase
D. DNA helicase
Answer: A
19. What is part B?




A. Codon
B. Anticodon
C. Amino acid
D. Ribosome
Answer: C

20. State suitable answer for one of the key functions of DNA.
A. Protein synthesis
B. Genetic information storage
C. Genetic information transfer
D. Amino acid delivery
Answer: B
21. Which of these can act as a vector to introduce new genes into a cell?
A. PCR
B. GMO
C. Plasmids
D. Clones
Answer: C
22. What shape of chromosomes are possessed in true-nucleus organisms?
A. Circular chromosomes
B. Long liner chromosomes
C. Square shape chromosomes
D. Oval shape chromosomes
Answer: B

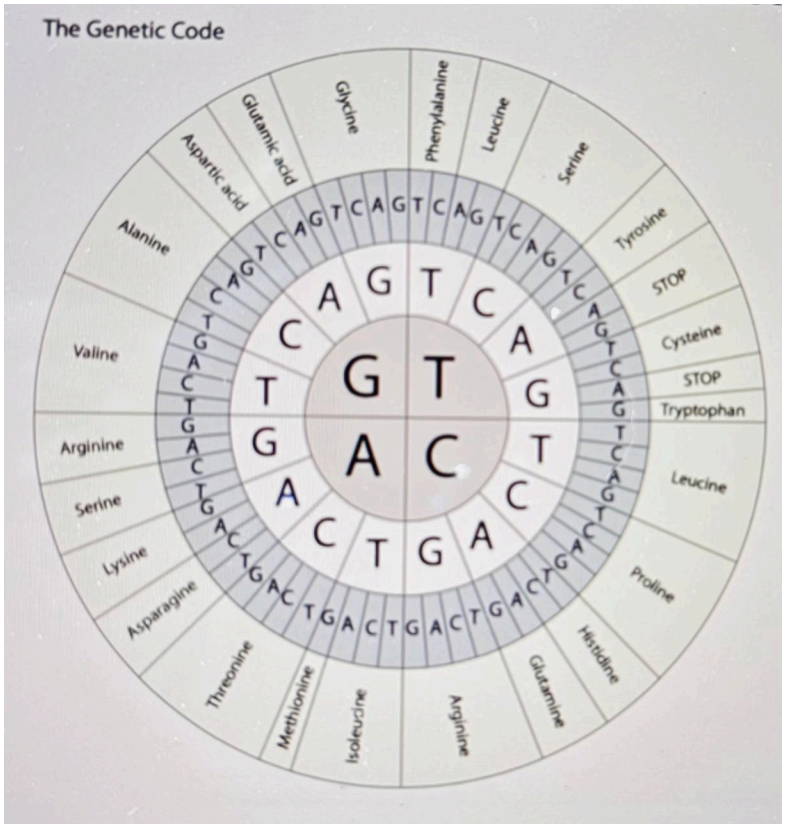
23. Which technique allows for the study of the enzymatic activity of proteins?
A. expression cloning
B. polymerase change reaction
C. DNA sequencing
D. gel electrophoresis
Answer: A
24. In prokaryotic DNA replication, how many times rapid in one second than eukaryotic DNA replication?
A. 10 times
B. 200 times
C. 100 times

D. 20 times
Answer: D
25. Which of the following statements is true about the casparian strip?
A. Prevents movement of water and solutes in apoplast
B. Prevents movement of water and solutes in symplast
C. It is located in the plasma membrane
D. It is located in the cell wall
Answer: A
26. Which type of cell in the phloem tissue has a nucleus?
A. Companion cells
B. Parenchyma cells
C. Phloem fibre
D. Sieve tube
Answer: A
27. What is the apoplastic pathway usually faster than the symplastic pathway?
A. It moves through cytoplasm
B. It uses active transport
C. It avoids cell membranes and travels through cell walls
D. It requires ATP
Answer: C
28. Why does transpiration increase in windy conditions?
A. Leaves absorb more carbon dioxide
B. Stomata close in wind
C. Xylem vessels become wider
D. Water potential gradient becomes steeper
Answer: D
29. What happens to a plant if the rate of transpiration is higher than water uptake?
A. Root pressure increases
B. The plant wilts due to loss of turgidity
C. Photosynthesis speeds up
D. The plant opens all stomata
Answer: B
30. The dissolve mineral ions are transported in the _____.
A. Xylem tissue
B. Phloem tissue
C. Phloem fibre
D. Sieve tube
Answer: A

SECTION B (10 x 2 = 20 MARKS)

1. A company wants to set up a system to grow crops in a location with poor soil quality. They choose

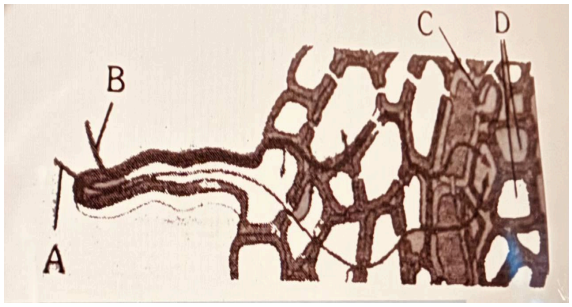
hydroponics as the growing method. Which of the following is a key advantage of using hydroponic
culture?
A. It requires large amounts of land to be effective
B. It relies entirely on soil quality to thrive
C. It allows crops to grow without soil, using a nutrient-rich water solution
D. It uses traditional farming methods to improve soil quality
Answer: C
2. Why spider silk is used in medicine for human?
a. Nerve and tissue regeneration
b. Remarkable combination of less tensile strength
c. Excellent biocompatibility and biodegradability
A. (a) and (b)
B. (a) and (c)
C. (b) and (c)
D. (a), (b) and (c)
Answer: B
3. Here are some of genetic technology:
(i). Gel electrophoresis
(ii). DNA sequencing
(iii). Expression cloning
Here are functions and applications of some genetic technology:
(a). genome mapping, diagnosing diseases
(b). gene identification and isolation, disease study
(c ). protein separation, DNA/RNA analysis
Matching genetic technology and functions and applications of genetic technology.
A. (i) – (c), (ii) – (a), (iii) – (b)
B. (i) – (b), (ii) – (c), (iii) – (a)
C. (i) – (a), (ii) – (b), (iii) – (c)
D. (i) – (a), (ii) – (c), (iii) – (b)
Answer: A
4. The antisense strand of a DNA molecule is given: AGTCATGTG
Use the following diagram to identify the amino acids from the mRNA sequence. The third amino acid is




AGTCATGTG
A. Arginine
B. Leucine
C. Glutamine
D. Histidine
Answer: D

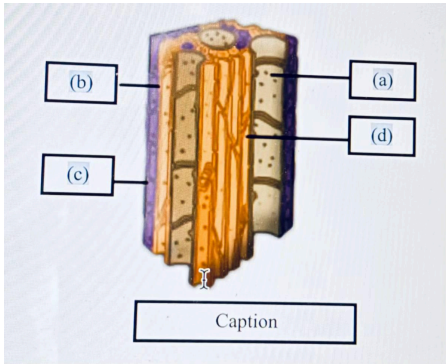
5. Step by step need to create golden rice:
A. Gene selection DNA construction Gene insertion Regeneration Selection Testing Approval
B. Gene insertion Gene selection Regeneration DNA construction Selection Approval Testing
C. Gene selection DNA construction Gene insertion Selection Regeneration Testing Approval
D. DNA construction Gene selection Gene insertion Testing Selection Approval Testing
Answer: C
6. RNA processing in prokaryotes and eukaryotes:
(i). location in prokaryotes – nucleus, location in eukaryotes – cytoplasm
(ii). splicing in eukaryotes – common, splicing in prokaryotes – rare or absent
(iii). capping in eukaryotes – no, capping in prokaryotes - yes
(iv). stability in prokaryotes- very low, stability in eukaryotes – high
A. (i), (ii), (iii) and (iv)
B. (ii), (iv)
C. (ii), (iii) and (iv)
D. (i), (iii) and (iv)
Answer: B

7. Here are differences between DNA and RNA:
(a). replication – DNA self-replicated while RNA is synthesized from DNA via transcription.
(b) Types – RNA exists in multiple forms while DNA has one main form
(c ). Length – DNA molecules are very long while RNA molecules are much shorter
(d). location – RNA is found in the nucleus, cytoplasm and ribosomes whereas DNA is primarily found
in nucleus and mitochondria
A. (a), (b) and (c)
B. (a), (b) and (d)
C. (a), (b) and (d)
D. (a), (b), (c) and (d)
Answer: D
8. The given diagram indicates routes of transport of water and minerals from the soil through the root.
Identify A, B, C and D.




A. A - Symplastic, B - Apoplastic, C - Casparian trip, D - Xylem vessels
B. A - Apoplastic, B - Symplastic, C - Cellulosic trip, D - Xylem vessels
C. A - Symplastic, B - Apoplastic, C - Casparian trip, D - Phloem vessels
D. A - Apoplastic, B - Symplastic, C - Cellulosic trip, D - Phloem vessels
Answer: A

9. Look at the Picture. Fill (a), (b), (c), (d) and caption.



A. (a) Xylem vessels, (b) Xylem fibers, (c) Phloem parenchyma, (d) Companion cells, Caption - Xylem
B. (a) Phloem vessels, (b) Tracheid, (c) Companion cells, (d) Xylem parenchyma, Caption - Phloem
C. (a) Xylem vessels, (b) Xylem fibers, (c) ) Xylem parenchyma, (d) Tracheid, Caption – Xylem
D. (a) Phloem Fibers, (b) Phloem parenchyma, (c) Sieve plate, (d) Phloem vessels, Caption – Phloem
Answer: C
10. A farmer wants to increase sugar content in fruits. According to the chapter, which practice would
MOST LIKELY achieve this?
A. Growing fruits in very humid conditions
B. Watering excessively to prevent transpiration
C. Closing stomata manually using chemicals
D. Allowing sufficient transpiration
Answer: D